Bank Customer Churn Analysis

& Retention Strategies

From Customer Insights to Retention Opportunities

Data Analysis & Generative AI
Final Project – The Hebrew University

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Bank Customer Churn Analysis & Retention Strategies

Business Challenge

Customer Churn can impact bank's revenue, customer relationships, and long-term growth.

Project Objective

Analyze historical churn patterns and identify opportunities for proactive customer retention.

Our Analytical Focus

Understand
Identify churn patterns and key customer characteristics.

→

Analyze
Explore churn trends, exit reasons and financial impact

→

Act
Prioritize higher-risk segments and recommend retention actions.

EDA & Data Enrichment

- Explored data structure and distributions across 10,000 customer records
- Validated data types, missing values, and duplicates
- Examined descriptive statistics for numerical variables
- Identified statistical outliers using the IQR method
- Created analytical groupings to support dashboard visualization

Synthetic Data Enrichment

Account Activation Date	Generated in alignment with customer tenure to support time-based analysis
Exit Date	Assigned to churned customers to analyze exit trends during 2023
Exit Reason	Assigned one of five predefined reasons to support churn-driver analysis and retention recommendations

Business Questions & Key Performance Indicators

Churn Overview & Segmentation

Where is the customer churn more significant?

Understand overall churn and identify customer characteristics associated with higher churn.

KPIs: Total Customers | Churned Customers | Churn Rate | Estimated Annual Revenue Loss

Churn Trends & Exit Reasons

How does churn evolve, and what drives customer exits?

Explore exit trends, reasons for leaving, and difference across customer segments.

KPIs: Exits Over Time| Exit Reason Distribution| Churn Rate by Customer Characteristics

Risk & Retention Opportunities

Which customer segments should be prioritized for retention?

Identify higher-risk segments and estimate their potential financial exposure.

KPIs: Historical Churn Rate| Adjusted Risk Level | Revenue at Risk | Expected Annual Revenue Loss

Assumptions & Methodology

Business Assumptions

- Annual revenue assumption: **4% of customer balance**
- Historical churn rates used to estimated potential future exposure
- Synthetic dates and exit reasons used for exploratory analysis

Risk Methodology

- Groups defined by age, geography, number of products and activity status
- Adjusted Risk used to stabilize small-group churn estimates
- Risk categories: Low (below 20%), Medium (20-40%), High (40%+)

Tools

- **Python (Pandas, NumPy, Matplotlib, Plotly):** Data cleaning, enrichments, analysis and visualization
- **Power BI (DAX):** Interactive dashboards, KPIs and data visualization.

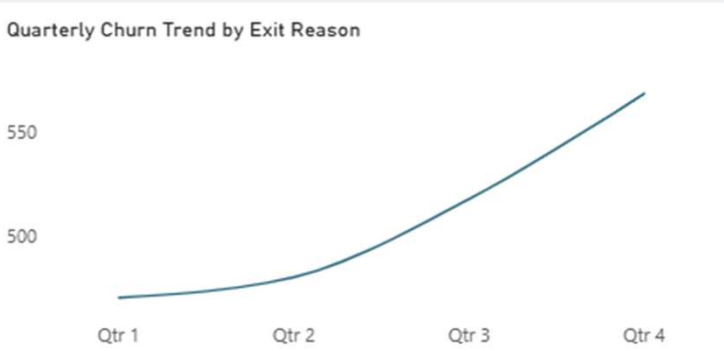
Customer Churn Overview & Segmentation

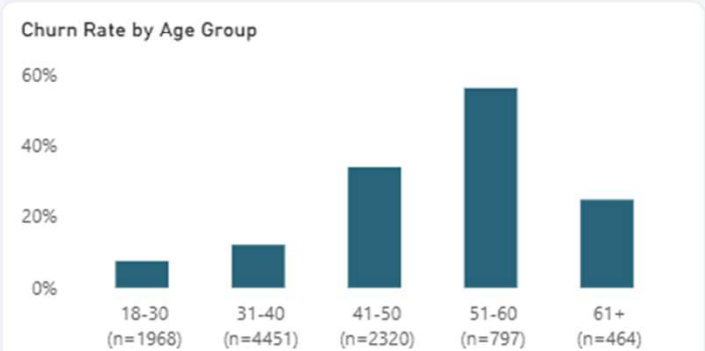
Number of Products: All

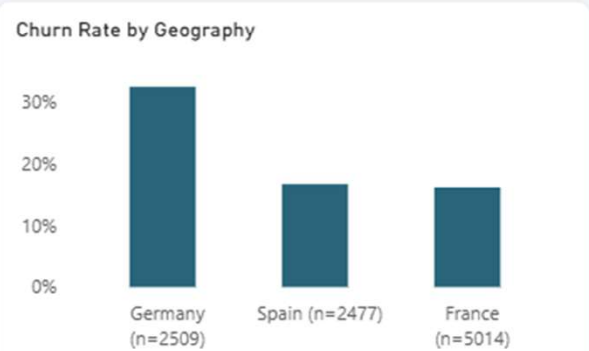
Age Group: All

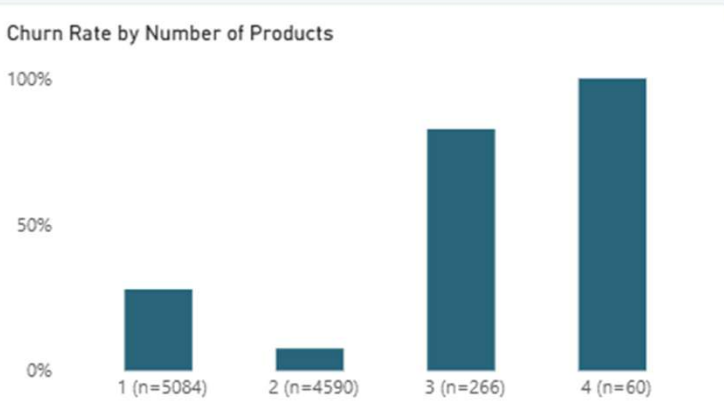
Geography: All

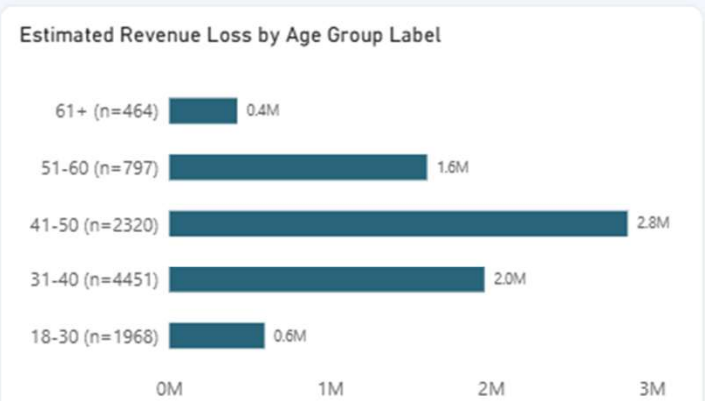
Total Customers	Churned Customers	Churn Rate	Estimated Revenue Loss
10,000	2,037	20.37%	7.42M

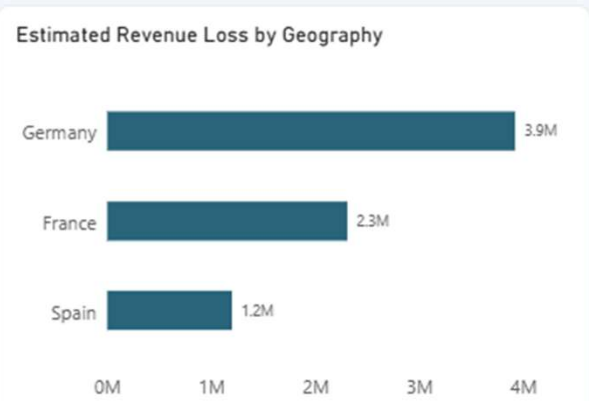












Key Findings – Customer Churn Overview

- The overall customer churn rate is **20.37%**
- Customer exits increased throughout the year, reaching their highest level in **Q4**.
- Most churned customers held only 1 product, while customer with 3-4 products showed the highest churn rates
- Customers aged 41-50 account for the largest number of exists, while the highest churn rate is observed among customers aged 51-60 (**56%**)
- Germany has the highest churn rate (**32.4%**) and accounts for approximately **53%** of the estimated annual revenue loss.

Customer Churn Trends & Exit Reasons

Number of Products: All

Age Group: All

Geography: All

Active Customers

churn rate

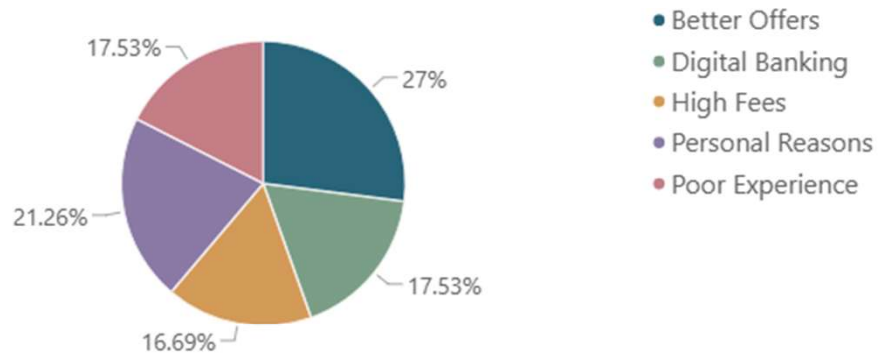
14%

Inactive Customers

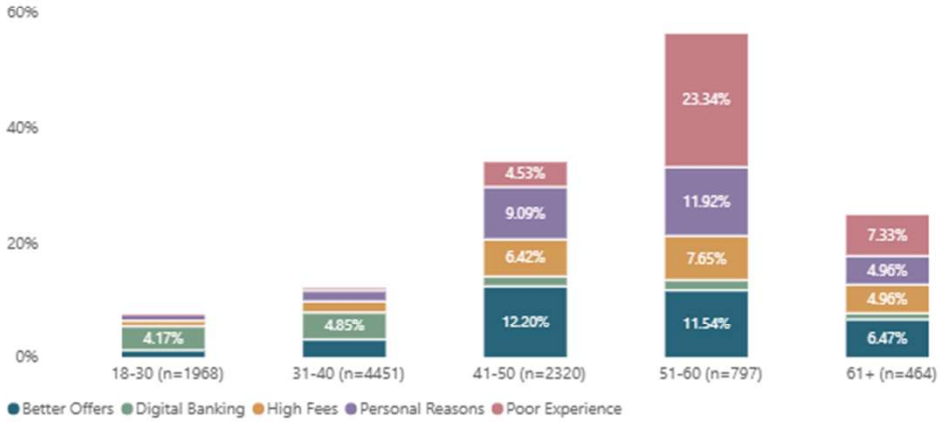
churn rate

27%

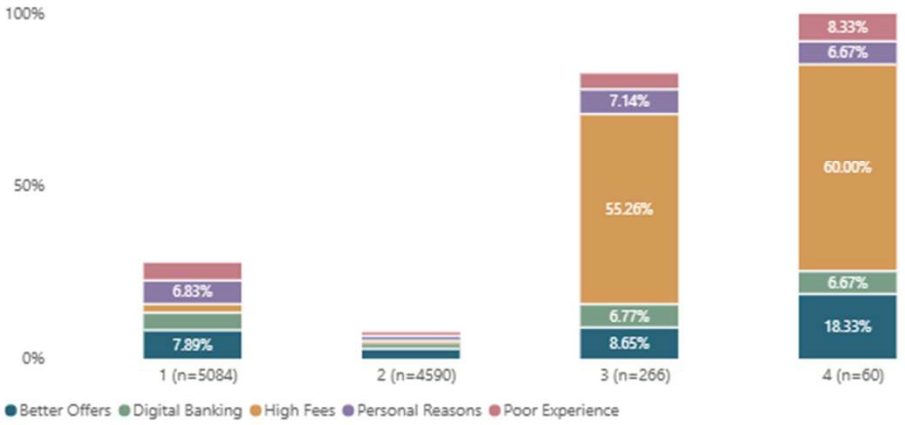
Distribution of Churn by Exit Reason



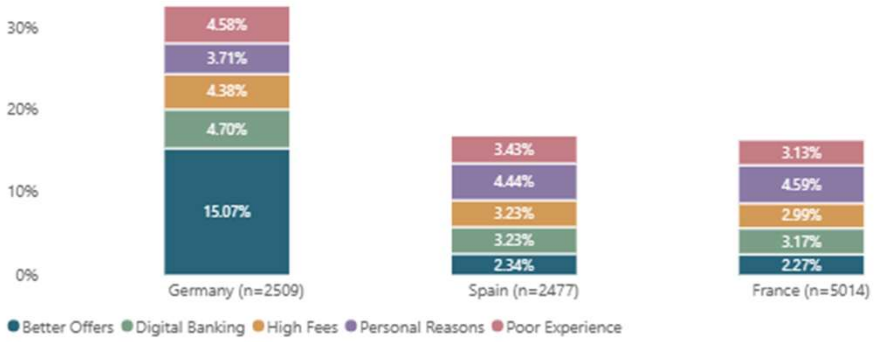
Churn Rate by Age Group



Churn Rate by Number of Products



Churn Rate by Geography



Key Findings & Recommendations – Churn Trends & Exit Reasons

Key Findings

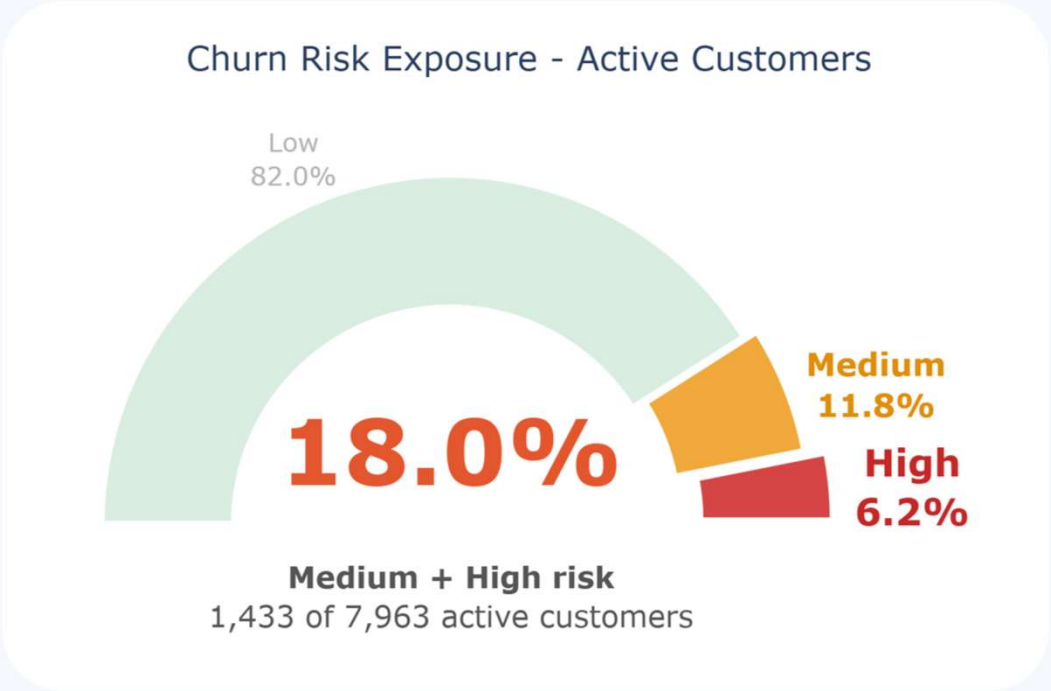
- **Competitive Offers:** Better Offers is the leading exit reason (27% of churned customers), particularly in Germany.
- **Age Differences:** Leading exit reasons vary across age groups, with Better Offers ranking first among customers aged 41-50.
- **Customer Activity:** Inactive customers have nearly twice the churn rate of active customers (27% vs. 14%)
- **Product Ownership:** Customers holding 3-4 products show exceptionally high churn rates (83-100%), although these segments are relatively small.

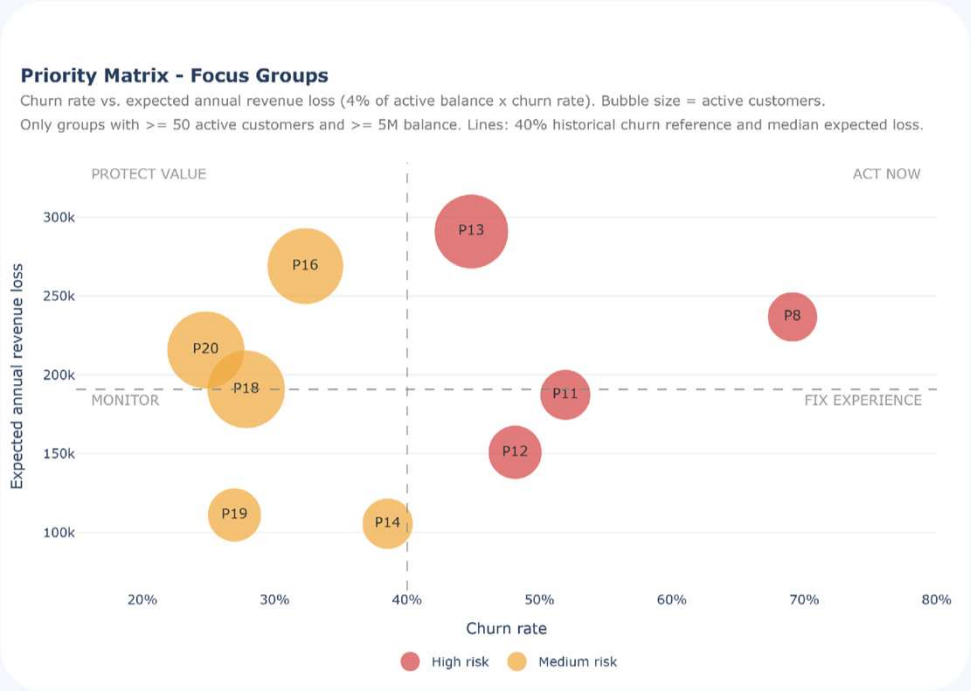
Recommendations

- **Competitive Offers:** Review competitors' offers and adjust pricing and benefits, particularly in Germany.
- **Product Adoption:** Test whether introducing a second product reduces churn among single product customers.
- **Customer Experience:** Explore digital improvements for younger customers (18-40) and service improvements for older customers (51+).
- **Fees:** Investigate whether fees contribute to high churn among customers holding 3-4 products.
- **Customer Engagement :** Asses whether inactivity predicts churn and test targeted re-engagement initiatives.

Churn Risk Overview

HIGH-RISK CUSTOMERS	BALANCE AT RISK	ESTIMATED ANNUAL REVENUE AT RISK
496	**51.03M**	**2.04M**
6.2% of active customers	High-risk customer balances	Based on 4% annual revenue assumption





Key Findings – Churn Risk Overview

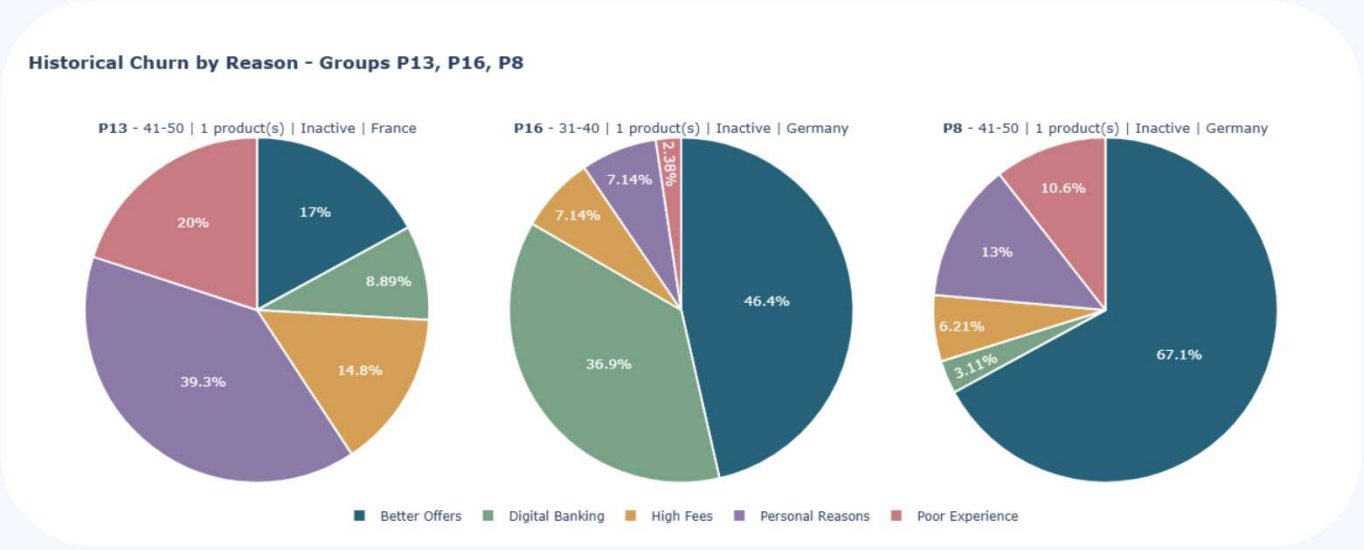
- **18%** of remaining customers (1,433) belong to **medium** or **high-risk** segments, including **496 high-risk customers**.
- High-risk customers hold **51.03M** in account balances, equivalent to **2.04M** in estimated annual revenue at risk, based on the 4% assumption.
- **P13** (France, ages 41-50, one product, inactive) has the highest estimated annual revenue loss among prioritized segments (**291K**).
- **P16** (Germany, ages 31-40, one product, inactive) ranks second in expected annual revenue loss (**296K**).
- **P8** (Germany, ages 41-50, one product, inactive) has the highest historical churn rate among the prioritized segments (**69%**).

Retention Priority Groups

Rank	Group	Customer Profile	Active Customers	Historical Churn	Est. Annual Rev. Loss
1	P13	41-50 | 1 product | Inactive | France	166	44.9%	291.0K
2	P16	31-40 | 1 product | Inactive | Germany	176	32.3%	269.0K
3	P8	41-50 | 1 product | Inactive | Germany	72	**69.1%**	236.8K
4	P20	31-40 | 1 product | Active | Germany	182	24.8%	215.8K
5	P18	41-50 | 1 product | Active | France	184	27.8%	190.8K
6	P11	41-50 | 1 product | Active | Germany	74	**51.9%**	187.3K
7	P12	41-50 | 1 product | Inactive | Spain	84	48.1%	150.9K
8	P19	18-30 | 1 product | Inactive | Germany	84	27.0%	111.1K
9	P14	51-60 | 1 product | Active | France	75	38.5%	105.6K

ADDITIONAL RISK INSIGHT

3-product customers: 82.7% historical churn; only 46 remain. This segment requires further investigation.



Retention Recommendations

#1 P13 · 41-50 | Inactive | Single Product | France ACT NOW

45% historical churn, 291K expected revenue loss

Why: Personal reasons 39% (x1.8 vs bank)

Action: remote / digital service + a relationship-manager retention call

#2 P16 · 31-40 | Inactive | Single Product | Germany PROTECT VALUE

32% historical churn, 269K expected revenue loss

Why: Better offers 46% (x1.7 vs bank) digital banking 37% (x2.1 vs. bank)

Action: Loyalty rate before exit + app onboarding with digital perks

#3 P8 · 41-50 | Inactive | Single Product | Germany FIX EXPERIENCE

69% historical churn, 237K expected revenue loss

Why: Better offers 67% (x2.5 vs bank)

Action: competitive rate / loyalty offer, prioritizing the 50 remaining customers with balances of 100K+

All three: inactive, single-product customers

Action: re-engagement campaign - personal call and usage incentive

Thank you

& Questions

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